

Object Oriented Programming

Assignment no: 2

Date: 24-Dec-2024

Instructor: Nimra Abbas

Total Marks: 40

Deadline: 30-Dec-2024

Name:

Roll no:

Rules:

- Submit your code files (.cpp) files in class maximum by 30-Dec-2024.
- Give proper comments where required.
- Plagiarism will lead to zero marks.
- No late submissions are allowed. You can only submit your task via google classroom.

Question 1:

[6]

Write a program in C++ to implement a class named “MyString” which will use char arrays as its private member. The class should have the following functions:

- a) Find the length of the string
- b) Concatenate two strings
- c) Make a sub-string
- d) Search a word in a string
- e) Reverse the string

Question 2:

[10]

Implement a class ComboLock that works like the combination lock in a gym locker, as shown here. The lock is constructed with a combination—of three numbers between 0 and 39. The reset method resets the dial so that it points to 0. The turnLeft and turnRight methods turn the dial by a given number of ticks to the left or right. The open method attempts to open the lock. The lock opens if the user first turns it right to the first number in the combination, then left to the second, and then right to the third.

```
class ComboLock :
ComboLock( int secret1, int secret2, int, secret3)
void reset() ;
void turnLeft(ticks) ;
...
void turnRight( ticks) ;
...
bool open() ;
```

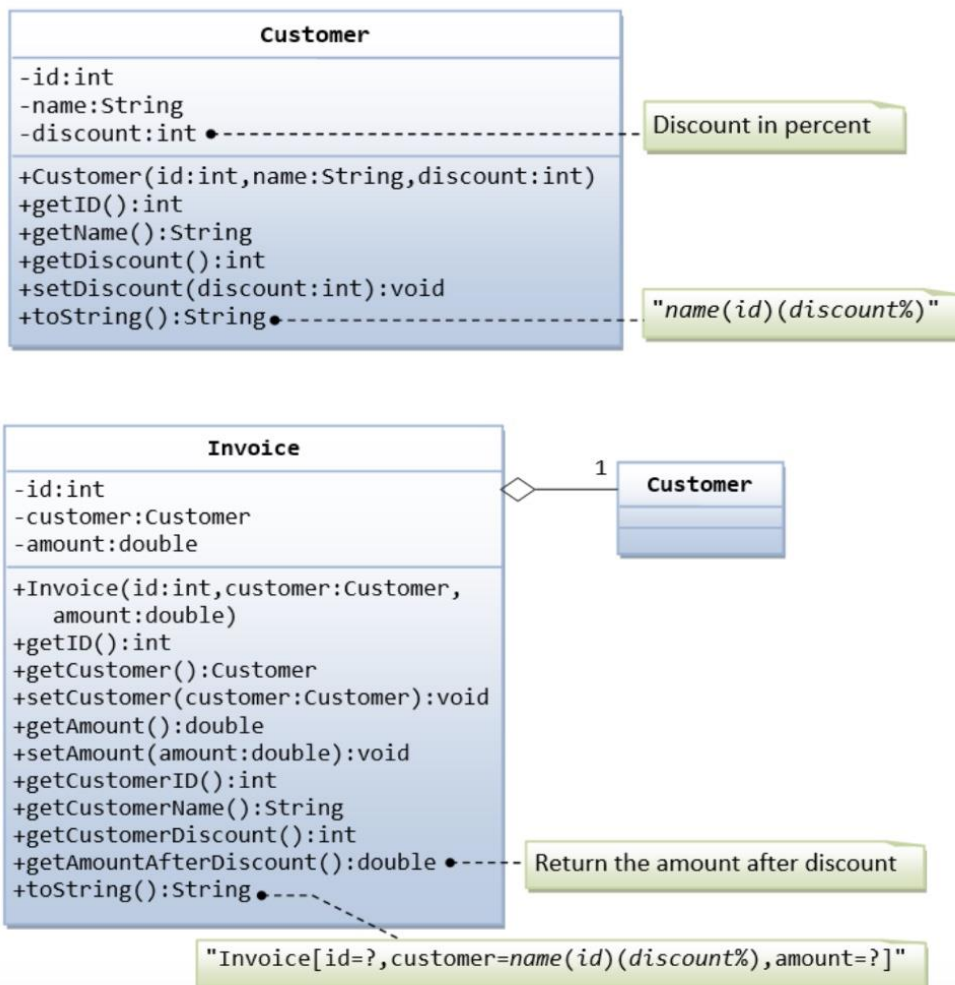
Question 3:

[10]

A university department consists of multiple teachers and offers various courses. Each course is taught by a teacher, and a teacher can teach multiple courses. Design a program to model this relationship using aggregation, where the Teacher class has a list of Course objects. Create classes Teacher and Course to represent teachers and courses, respectively. The Teacher class should contain attributes like id, name, and a list of courses they teach. The Course class should have attributes like id, title, and teacherId. Implement methods to add courses to a teacher and to display all courses taught by a teacher.

Question 4: Implement the given :

[6 + 8]



b) Implement a superclass Appointment and subclasses Onetime, Daily, and Monthly. An appointment has a description (for example, “see the dentist”) and a date. Write a method occursOn(year, month, day) that checks whether the appointment occurs on that date. For example, for a monthly appointment, you must check whether the day of the month matches. Then fill a list of Appointment objects with a mixture of appointments. Have the user enter a date and print out all appointments that occur on that date.

***** The End *****